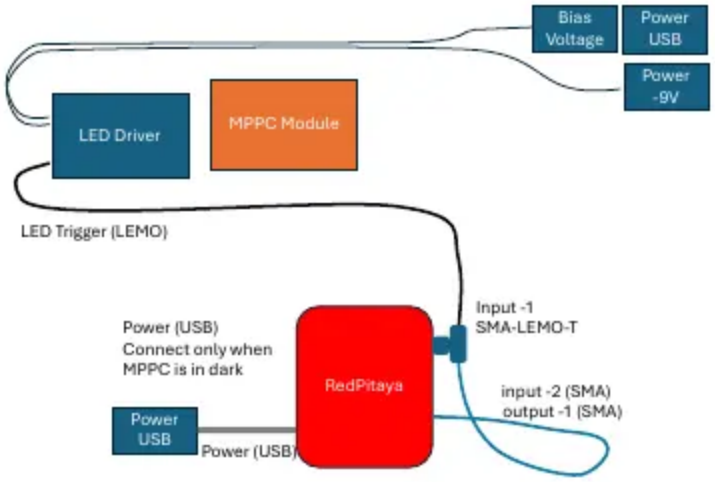
Photon Counting Experiment

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An activity of photon counting detection.

Introduction

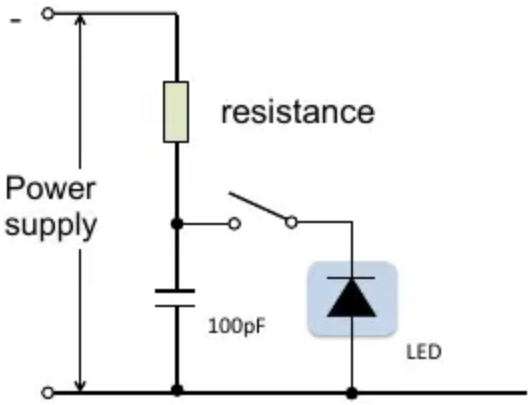
This activity tries to achieve photon counting from a weak light source where photons are emited. If we have a highly sensitive detector (SiPM) with enough gain we can collect some of the photons.

Setup experiment



DAQ processing are gonna be in Red Pitaya and data lecture and analysis in a conventional laptop connected via WiFi.

Light Pulser



Light was emitted by blue LED ($\lambda = 465\ \mathrm{nm}$) where all charge is given by capacitor $c = 100\ \mathrm{pF}$. Exceded voltage allows excedent of charge that leads to LED bright.

✓ Necessary energy of a blue photon.

It will require

$$E = h\nu = \frac{hc}{\lambda} = \frac{2\pi\hbar c}{\lambda} = \frac{2\pi(197\,\mathrm{eV}\cdot\mathrm{nm})}{465\,\mathrm{nm}} = 2.67\,\mathrm{eV}$$

This is the minimum voltage, but it will be barely visible.

In this exercise, we will need 3.2 V due to resistence and so on. For example, if 4.8 V is applied, the excess voltage is 1.6 V. Then charge is $Q = CV = (100\ \mathrm{pF})(1.6\ \mathrm{V}) = 160\,\mathrm{pC}$.

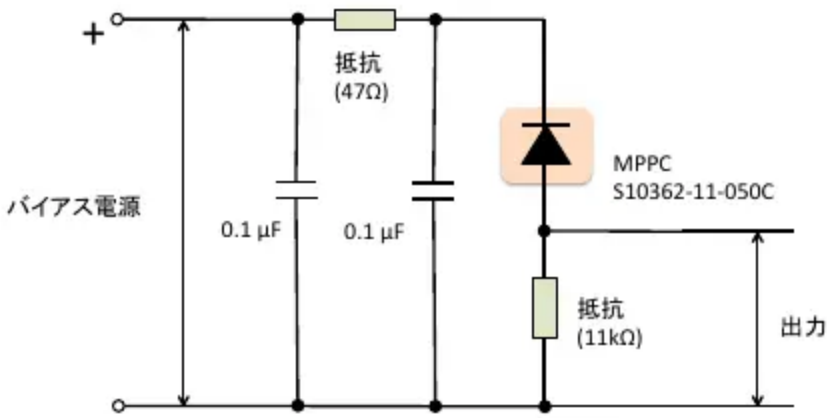
✓ 160 pC of electrons flow through the LED. How many electrons is that?

$$N_e = \frac{C}{e} = \frac{160\,\mathrm{pF}}{1.6\times10^{-19}\,\mathrm{C}} = 10^9\ \mathrm{electrons}$$

It's hard to predict how much photons will emitt (LED properties, internal resistance, etc.).

Pulses were seen on a digital oscilloscope *MPPC Scope*. It was set the trigger rate to 1000 Hz for a bias voltage of -11 to -13 V. This will emitt even less light for photon counting. For emitting light it's necessary to get

MPPC (Multi-pixel Photon Counter)

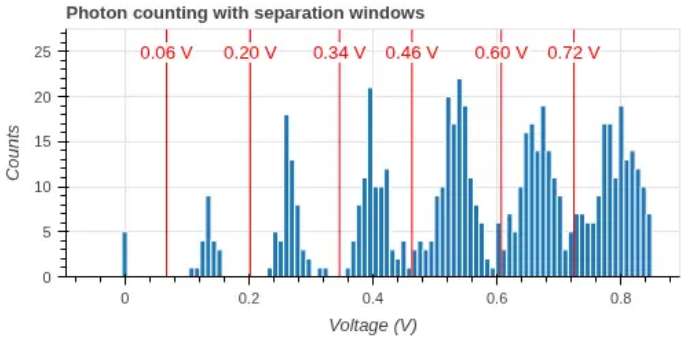


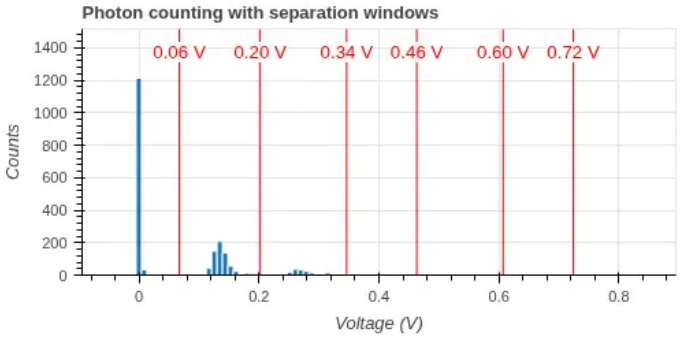
- Composed of many pixels, each of one an avalanche photodiode.
- Operates in Geiger mode.
- S10362-11-050C has 400 pixels.
- Don't operate under bright light.
- Reverse voltage of 70 V→ dark current.

MPPC Scope for this dark current (100 kcps) takes in a coarse frame (4 μs) around 0.4 counts. That is approximately 0-2 pulses expected. *This is a signal to continue.*

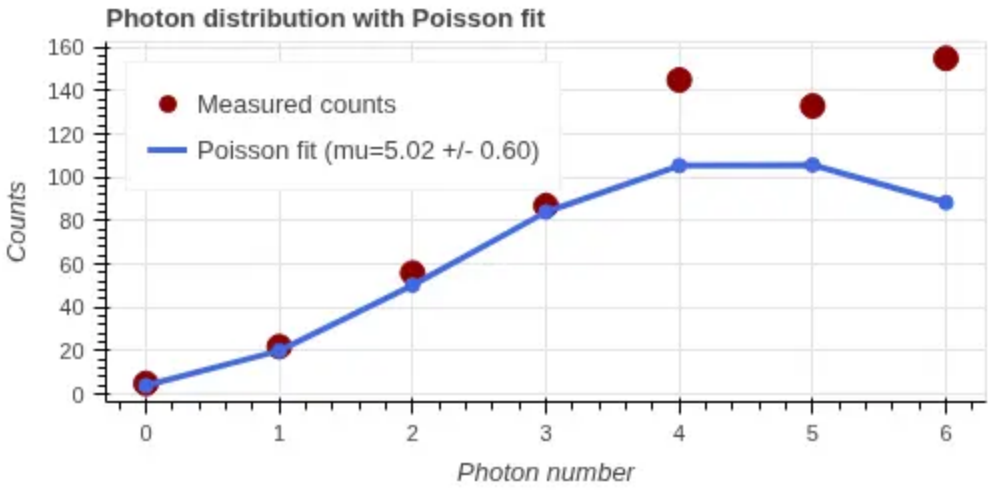
Histogram

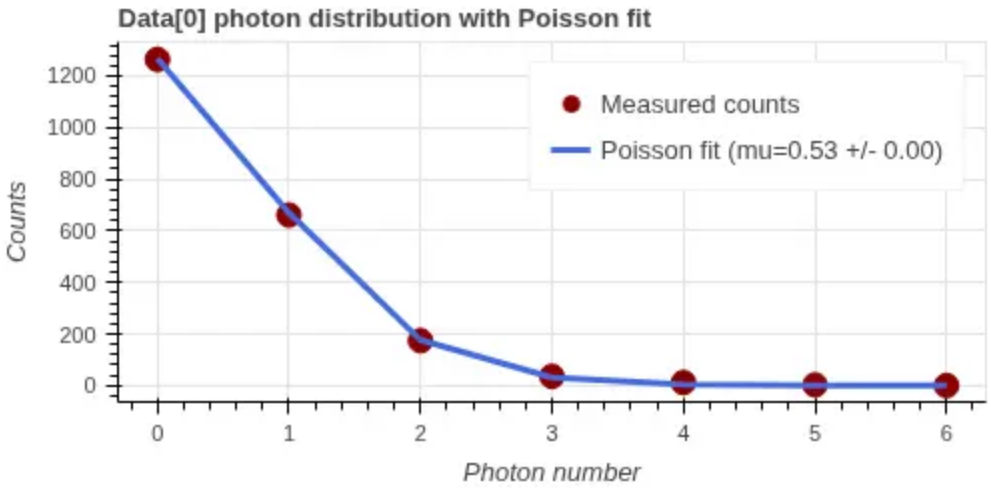
In our case, there were **two measurements**





where the results are





Accurated analysis on notebook.